SEMESTER: III						
Course Code	:	MCA261IA	**FULL STACK APPLICATION DEVELOPMENT**	CIE Marks	:	100 +50
Credits L-T-P	:	3-0-1	*(Theory & Practice)*	SEE Marks	:	100 + 50
Hours	:	45L + 30P	*(Professional Core Course with Integrated Lab)*	SEE Duration	:	3 Hours

UNIT - I	**9 Hours**

Introduction to Full Stack Development: Understanding Full Stack Development, Technologies associated with Full Stack

Introduction to the MERN stack: Introduction, The MVC architectural Pattern, MERN Components React, Node.js, Express, MongoDB, Advantages of MERN, Isomorphic;

Node.js - event-driven programming, JavaScript closures Node modules, Common JS modules, Node.js core modules, Node.js third-party modules, Node.js file modules Developing Node.js web application, connecting and mounting middleware;

Building Express Web Application - Introduction to Express, Installing Express, Creating your first Express application, The application, request and response objects- The application object, The request object, The response object, External middleware

UNIT - II	**9 Hours**

Understanding React and Web Server: Server setup, NVM Node JS, Project, NPM, Express, Build time JSX compilation- Separate Script File, Transform, Automate, React Library, React Components-React classes, Composing components, passing data- using properties, property validation, using children's Dynamic composition;

Understanding React State: React State: Setting state, Event handling, communicating from child to parent, Stateless components, Designing Components-state vs props, component hierarchy communication, Stateless components

UNIT - III	**9 Hours**

Introduction to MongoDB: Introduction to NoSQL, Introducing MongoDB, MongoDB sharding,

MongoDB CRUD operations-Creating a new document, Creating a document using insert(), Creating a document using update(), Creating a document using save();

Introduction to Mongoose-Introducing Mongoose, Connecting to MongoDB, Understanding Mongoose schemas, Creating the user schema and model, Registering the User model, Creating new users using save(), Finding multiple user documents using find(), Reading a single user document using find One(), Updating an existing user document Deleting an existing user document;

Extending your Mongoose schema- Defining default values, Using schema modifiers, Predefined modifiers, Custom setter modifiers, Custom getter modifiers

UNIT - IV	**9 Hours**

Building RESTful APIs and Mongo DB: REST, HTTP method as Actions, JSON, Express, Routing Hander function , Request Object, Response objects, Middleware, The list API , The create API, Using the LIST API, Using the Create API, Error Handling

UNIT - V	**9 Hours**

Working with React Router and forms: Routing Techniques, Simple Routing, Route parameters, Route Query String, Programmatic Routing, Nested Routes, Browser history, Forms, Filter form, Get API, Edit page, UI Components, update API, Delete API.

LABORATORY

1.Demonstrate Node. Js Application to perform CRUD operation for online Book Cart
2. Write a node.js program using Express framework to accept user name, Branch, Semester, from web page and display the information as below
   a) Handle both get and post methods
   b) Branch should be underlined
   c) Name should be in bold face.
3. Design a resume of a job aspirant using React components like Classes and Functions. Style the resume by applying CSS
4. Build student registration portal using Entities like component, state and props
5. Design and implement a React Form that collects user input for name, email, and password. Validate the form using Regular Expression.
6. Deploy connectivity between React and Node Application for Inventory Management system
7. Develop a MongoDB query for comparison selectors, Logical Selectors for Company database
8. Execute aggregation pipeline and its operation to illustrate text search on catalog data collection
9. Design an employee Management system using RESTFULL APIs in React
10. Create a React application using react-router-dom with multiple pages (Home, About, Contact).

Course Outcomes:

After going through this course the student will be able to:

CO1	:	Demonstrate the core concepts of the Model-View-Controller (MVC) architecture and its role in structuring web applications
CO2	:	Apply the MVC pattern in building applications using the MERN stack components
CO3	:	Design and develop full-stack web applications using the MERN stack
CO4	:	Analyze the structure and functionality of web applications by implementing the MVC framework using the MERN stack

Reference Books

1.Subramanian, V. “Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node” Apress. 2nd edition, 2019, ISBN: 9781484243904

2.Hoque, S. “Full-Stack React Projects: Learn MERN Stack Development by Building Modern Web Apps Using MongoDB, Express, React, and Node.js”, Packt, 2nd Edition, 2022 Publishing. ISBN: 9781801070636

3.Ackermann, P. “Full Stack Web Development: The Comprehensive Guide, Packt publication, Rheinwerk Computing, 2023, ISBN: 9783969108830

4.Osmani, A. , ”Learning JavaScript Design Patterns: A JavaScript and React Developer’s Guide” 2nd Edition, 2023, O’Reilly Media. ISBN: 9781098139872

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)

CIE will consist of TWO Quizzes (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[20 (Q) + 40 (T) + 40 (EL) = 100 marks]**

Sl.No.	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted & each Quiz will be evaluated for 10 marks, and Final Quiz marks adding up to 20 marks. **THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom’s Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO TESTS** will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST and Quiz to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS**.	**40**
	CIE THEORY TOTAL	**100**

RUBRIC FOR CONTINUOUS INTERNAL EVALUATION (CIE-Lab)

Q.NO.	CONTENTS	MARKS
1	Conduction of the Experiments & Lab Record	40
2	Lab Test	10
	CIE LAB TOTAL	**50**
	MAXIMUM MARKS FOR THE CIE	**150**
RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	SEE THEORY TOTAL	**100**
RUBRIC FOR SEMESTER END EXAMINATION (SEE-Lab)		
Q.NO.	**CONTENTS**	**MARKS**
1	Design and Development of Project	20
2	Presentation of working model/simulation results/prototype building	20
3	Viva voce	10
	SEE LAB TOTAL	**50**
	MAXIMUM MARKS FOR THE SEE	**150**

SEMESTER: III						
Course Code	:	MCA262IA	**DEVOPS AUTOMATION**	CIE Marks	:	100+50
Credits L-T-P	:	3-0-1	*(Theory & Practice)*	SEE Marks	:	100+50
Hours	:	45L + 30P	*(Professional Core Course with Integrated Lab)*	SEE Duration	:	03 Hours

UNIT - I	**09 Hours**
The DevOps Culture: Getting started with DevOps **Docker Fundamentals:** Discovering Docker, What and why of Docker, Building a Docker Application. Understanding Docker - Docker's Architecture, The Docker Daemon, The Docker Client, Docker Registries, The Docker Hub	
UNIT - II	**09 Hours**
Docker and Development: Using Docker as a lightweight Virtual Machine - From VM to Container, Saving and restoring your work, Environments as processes, Building Images, Running Containers	
UNIT - III	**09 Hours**
Docker and DevOps: Continuous Integration - Docker Hub automated builds, containerizing your CI process - Running the Jenkins master within a Docker container. Continuous delivery - Interacting with other teams in the CD pipeline, facilitating deployment of Docker images, Configuring your images for environments, Upgrading running containers	
UNIT - IV	**09 Hours**
First steps with Docker and Kubernetes: Creating, running, and sharing a container image, Setting up a Kubernetes cluster - Running a Local Single-Node Kubernetes Cluster with Minikube, Setting up an alias and command-line completion for kubectl, Running the first app on Kubernetes - Deploying your Node.js app, accessing your web application, The logical parts of your system, Examining what nodes your app is running on, Introducing the Kubernetes dashboard	
UNIT - V	**09 Hours**
Pods: Introducing Pods, Creating pods from YAML or JSON descriptors, organizing pods with labels, Listing subsets of pods through label selectors, Annotating pods, Using namespaces to group resources, Stopping and removing pods	

LABORATORY

1. Build a Docker Container from a Custom Dockerfile
2. Develop a Multi-Stage Dockerfile for Container Orchestration.
3. Code a Dockerized Python Flask or Node.js Application
4. Integrate Git with Docker for Source-Controlled Application Builds
5. Demonstrate CI Integration by Running Jenkins in a Docker Container
6. Deploy an Automated Build Pipeline using Docker Hub
7. Deploy a Web Application to Kubernetes using Minikube
8. Create Kubernetes Pods using YAML Descriptors
9. Organize Kubernetes Pods Using Labels and Namespaces
10. Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

Course Outcomes:		
After going through this course the student will be able to:		
CO1	:	Understand the fundamentals of DevOps and demonstrate the use of Docker for container creation and management.
CO2	:	Build and run Docker containers to simulate virtual environments and enable consistent development workflows.
CO3	:	Develop automated CI/CD pipelines using Jenkins and Docker Hub to streamline integration and deployment processes.
CO4	:	Deploy containerized applications on Kubernetes and manage pods, labels, and namespaces for efficient workload orchestration

Reference Books

1.Gaurav Agarwal, "Modern DevOps Practices: Implement, secure, and manage applications on the public cloud by leveraging cutting-edge tools", Packt Publishing, 2nd Edition, 2024, ISBN–9781805121824

2. Ian Miell, Aidan Hobson Sayers, "Docker in Practice", Manning Publications, 2nd Edition, 2019, ISBN–9781617294808

3.Marko Lukša, "Kubernetes in Action", Manning Publications, 2nd Edition, 2018, ISBN–9781617293726

4. Brendan Burns, Joe Beda, and Kelsey Hightower, "Kubernetes: Up and Running", 2nd Edition, 2019, ISBN–978-1-492-04653-0

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)

CIE will consist of TWO Quizzes (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[20 (Q) + 40 (T) + 40 (EL) = 100 marks]**

Sl.No.	COMPONENTS	MARKS
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted & each Quiz will be evaluated for 10 marks, and Final Quiz marks adding up to 20 marks. **THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO TESTS** will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST and Quiz to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS**.	**40**
	CIE THEORY TOTAL	**100**

RUBRIC FOR CONTINUOUS INTERNAL EVALUATION (CIE-Lab)

Q.NO.	CONTENTS	MARKS
1	Conduction of the Experiments & Lab Record	40
2	Lab Test	10
	CIE LAB TOTAL	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	SEE THEORY TOTAL	**100**
RUBRIC FOR SEMESTER END EXAMINATION (SEE-Lab)		
Q.NO.	**CONTENTS**	**MARKS**
1	Design and Development of Project	20
2	Presentation of working model/simulation results/prototype building	20
3	Viva voce	10
	SEE LAB TOTAL	**50**
	MAXIMUM MARKS FOR THE SEE	**150**

SEMESTER: II						
Course Code	:	MCA263D1	**ADVANCED INTERNET OF THINGS**	CIE Marks	:	50
Credits L-T-P	:	3-0-0	*(Theory)*	SEE Marks	:	100
Hours	:	45L	*(Professional Elective Course) - D*	SEE Duration	:	3 Hours

UNIT - I	**9 Hours**

Advanced IoT Architecture, Ecosystem and protocols beyond MQTT

IoT layered architecture (perception, network, application),Smart objects and digital twins IoT platforms comparison (AWS IoT, Azure IoT, ThingsBoard, etc.), Design considerations: interoperability, scalability, latency, Case study: Industrial IoT vs Consumer IoT CoAP, XMPP, AMQP, DDS – overview and use cases, 6LoWPAN, Zigbee, BLE Mesh, Thread LoRaWAN architecture and use cases, Edge vs Fog vs Cloud communication

Interfacing protocols: I2C, SPI, UART, Modbus RTU/TCP

UNIT - II	**9 Hours**

IoT Security and Data Integrity

IoT attack surfaces and threat models, Secure communication (TLS, DTLS, VPNs) Authentication (OAuth2, API Keys, Tokenization), Secure firmware updates and boot mechanisms Privacy, identity, and data encryption techniques, Blockchain in IoT security (introduction level)

UNIT - III	**9 Hours**

Edge AI and ML in IoT

Edge AI platforms (NVIDIA Jetson, Coral, ESP32-S3),Data acquisition and feature extraction at the edge, On-device model inference (TensorFlow Lite, Edge Impulse), Real-time decision making using micro-models, Use cases: predictive maintenance, anomaly detection

UNIT - IV	**9 Hours**

IoT Data Management and Visualization and

Time-series databases (InfluxDB, TimescaleDB),Data ingestion pipelines (Node-RED, Apache NiFi), Dashboards (Grafana, ThingsBoard etc), Event triggers and notification systems Integration with cloud storage (AWS S3, Google Firebase etc)

UNIT - V	**9 Hours**

Advanced Application Development

IoT DevOps – containerization (Docker), CI/CD, OTA updates,Multi-node coordination and orchestration, Interoperability with external APIs and ERP systems, Use of REST/GraphQL APIs for remote access, Design and development of end-to-end IoT projects

Course Outcomes:

After going through this course the student will be able to:

CO1	:	Analyze the architecture and design considerations for advanced IoT systems, including industrial applications.
CO2	:	Compare and implement various IoT communication protocols beyond MQTT for efficient and secure data transmission.
CO3	:	Develop secure IoT applications incorporating encryption, authentication, and firmware integrity mechanisms and Deploy machine learning models on edge devices to enable local decision-making in IoT systems.
CO4	:	Design data ingestion, storage, and visualization pipelines for real-time monitoring and analytics and Integrate cloud and edge systems to create full-stack, intelligent, and secure IoT solutions.

Reference Books
1. Rolando Herrero, "Practical Internet of Things Networking: Understanding IoT Layered Architecture", January 2023, Springer DOI:10.1007/978-3-031-28443-4, ISBN: 978-3-031-28442-7, Northeastern University.
2. Daniel Situnayake, "AI at the Edge: Solving Real-World Problems with Embedded Machine Learning" ,Jenny Plunkett, O'Reilly Media, 1st Edition, 2023 , ISBN-13. 978-1098120207;
3. Smita Jain, Vasantha Lakshmi, Dr Rohini Srivathsa, "IoT and OT Security Handbook: Assess Risks, Manage Vulnerabilities, and Deploy Secure Systems" Packt Publishing,2023 ISBN 978-180461980.
4. ByRute C. Sofia, John Soldatos, Shaping the Future of IoT with Edge Intelligence,1st Edition 2024, River Publishers, DOIhttps://doi.org/10.1201/9781032632407
5. Sudip Misra, Chandana Roy , Anandarup Mukherjee, "Introduction to Industrial Internet of Things and Industry 4.0", CRC Press, 2021
URL resources
Node-RED Docs https://nodered.org/docs/ Mosquitto MQTT Broker https://mosquitto.org/ LoRaWAN Protocol Overview https://lora-alliance.org/lorawan-specification/ CoAP Protocol (RFC 7252) https://datatracker.ietf.org/doc/html/rfc7252 ESP32 Technical Reference Manual https://www.espressif.com/en/support/download/documents Edge Impulse (ML on Microcontrollers) https://docs.edgeimpulse.com/docs ThingsBoard IoT Platform https://thingsboard.io/docs/ Grafana Visualization Platform https://grafana.com/docs/grafana/latest/ TensorFlow Lite for Microcontrollers https://www.tensorflow.org/lite/microcontrollers OpenCV for Embedded Vision https://opencv.org/platforms/embedded/ OWASP IoT Top 10 Security Risks https://owasp.org/www-project-internet-of-things/

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of TWO Quizzes (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[20 (Q) + 40 (T) + 40 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. **TWO QUIZZES** will be conducted & each Quiz will be evaluated for 10 marks, and Final Quiz marks adding up to 20 marks. **THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**20**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO TESTS** will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST and Quiz to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS**.	**40**
	CIE THEORY TOTAL	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	SEE THEORY TOTAL	**100**

SEMESTER: III						
Course Code	:	MCA263D2	**DEEP LEARNING**	CIE Marks	:	100
Credits L-T-P	:	3-0-0	*(Theory)*	SEE Marks	:	100
Hours	:	45L	(Professional Elective Course) -D	SEE Duration	:	3 Hours

UNIT - I	**9 Hours**
Neural Networks: Introduction to NN, models of neuron and network architectures. Learning Processes: Different types of learning processes, Learning with and without teacher, Memory, statistical learning theory. **Single layer perceptron**: Adaptive filter problem, least mean square algorithm, learning rate, Learning rate, annealing techniques, perceptron and perceptron convergence theorem. **Multilayer Perceptron:** Back propagation algorithm, Sequential and batch modes of training, stopping criteria, XOR problem, and some numerical problems	
UNIT - II	**9 Hours**
Convolutional Neural Networks: Introduction, Historical Perspective and Biological Inspiration. **Basic Structure of a Convolutional Network**: Padding, Strides, Typical Settings, The ReLU Layer, Pooling, Fully Connected Layers, The Interleaving Between Layers, Local Response Normalization, Multiplications, Data Augmentation. **Training a Convolutional Network:** Back propagating Through Convolutions, Back propagation as Convolution with Inverted/Transposed Filter, Convolution/Back propagation as Matrix	
UNIT - III	**9 Hours**
Convolutional Architectures: AlexNet, ZFnet VGG, GoogLeNet, ResNet Effect of Depth, Pretrained Models. **Applications of CNN:** Content based image retrieval, Object Localization, Object Detection, Natural Language and sequence learning, and Video classification	
UNIT - IV	**9 Hours**
Recurrent Neural Networks: Introduction and expressiveness of RNN. Basic Structure of a RNN: Language Modeling Example of RNN, Generating a Language Sample, Back propagation Through Time, Bidirectional Recurrent Networks, Multilayer Recurrent Networks. Echo-State Networks, Long Short-Term Memory (LSTM), Gated Recurrent Units (GRUs) **Applications of Recurrent Neural Networks:** Automatic Image Captioning, Temporal Recommender Systems, Secondary Protein Structure Prediction, End-to-End Speech Recognition, Handwriting Recognition	
UNIT - V	**9 Hours**
Deep Reinforcement Learning: Introduction Stateless Algorithms: Multi-Armed Bandits: Naïve Algorithm, Greedy Algorithm, Upper Bounding Methods The Basic Framework of Reinforcement Learning: Challenges of Reinforcement Learning, Simple Reinforcement Learning for Tic-Tac-Toe, Role of Deep Learning and a Straw-Man Algorithm	

Course Outcomes:
After going through this course the student will be able to:

CO1	:	Apply basic concepts of neural network, its applications and various learning models
CO2	:	Analyse different Network Architectures, learning tasks, convolutional networks, and deep learning models
CO3	:	Assess neural networks model and learning techniques to solve problems related to society and industry
CO4	:	Demonstrate a prototype application developed using any NN tools and APIs

Reference Books
1. Simon Haykin, "Neural Networks – A Comprehensive Foundation", 2nd Edition, PHI, 2005.
2. Charu C Aggarwal, "Neural Networks and Deep learning: A Textbook", Springer International Publishing AG, ISBN 978-3-319-94462-3 ISBN 978-3-319-94463-0 (eBook), https://doi.org/10.1007/978-3-319-94463-0, 2018
3. Gunjan Goswami, S.K. Kataria & Sons, "Introduction to Artificial Neural Networks", 3rd edition, 2012, ISBN-13:978-9350142967.
4. Nikhil Buduma, "Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms", O'Reilly Publications, 1st Edition,2017, ISBN-13: 978-1491925614.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of TWO Quizzes (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[20 (Q) + 40 (T) + 40 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & each Quiz will be evaluated for 10 marks, and Final Quiz marks adding up to 20 marks. THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.	**20**
2.	**TESTS:** Students will be evaluated in a test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS**. **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: III						
Course Code	:	MCA263D3	**ADVANCED COMPUTER NETWORKS**	CIE Marks	:	100
Credits L-T-P	:	3-0-0	*(Theory)*	SEE Marks	:	100
Hours	:	45L	(Professional Elective Course) -D	SEE Duration	:	3 Hours
Faculty Coordinator:						

UNIT - I	**9 Hours**
Internet Protocol- Introduction, Error and Control Messages (ICMP): The Internet Control Message Protocol, Error Reporting vs Error Correction, Testing Destination Reachability and status, Echo Request and Reply Message Format. Classless and Subnet Address Extension (CIDR): Review of Relevant Facts, Proxy ARP, Subnet Addressing, Subnet Mask Representation, Broadcasting the Subnets, A Classless Addressing example	
UNIT - II	**9 Hours**
Wireless LANS and PANS – Fundamentals of WLAN's, 802.11 Standards, HIPERLAN Standard, Blue tooth specifications, Transport Protocol group, ZigBee Specification Wireless WANS and MANS – The Cellular Concept and Cellular Architecture- Capacity enhancement, Channel Allocation Algorithms	
UNIT - III	**9 Hours**
Mobile IP - Introduction, Mobility, Routing and Addressing, Mobile IP Characteristics, Overview of Mobile IP Operations, Mobile Addressing Details, Foreign Agent Discovery, Agent Registration, registration message format, communication with a foreign agent, datagram transmission and reception, two- crossing problem, communication with computers on the home network Private	
UNIT - IV	**9 Hours**
Advanced Internetwork Router Implementation: The Global Internet-Routing Areas, Inter domain Routing (BGP), IP Version 6(IPv6), Multiprotocol Label Switching (MPLS)-Destination Based forwarding, Explicit Routing, Network Interconnection- NAT, VPN- Introduction	
UNIT - V	**9 Hours**
SDN - Introduction, Centralized and Distributed Control and Data Planes- Introduction, Control plane, Data plane, Moving Information Between Planes, Distributed Control Planes, IP and MPLS, Convergence Time, Load Balancing, High Availability.	

Course Outcomes:
After going through this course the student will be able to:

CO1	:	Apply the advanced networking concepts
CO2	:	Identify various networking classifications in day to day computing
CO3	:	Analyze the importance of routing and congestion control principles
CO4	:	Access the different routing protocol methods in the networking support layers

Reference Books

1. James F. Kurose and Keith W. Ross," Computer Networking: A Top-Down Approach", Pearson Education, 8th edition, 2023, ISBN: 978-0136685208
2. William Stallings, "Wireless Communications & Networks", Pearson Education, 7th Edition, 2020, ISBN: 978-0134799085.
3. Patricia Marechal and Giovanni Chiola, "Software Defined Networking: Design and Deployment", Springer International Publishing, First Edition, 2023,ISBN: 978-3031234567.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of TWO Quizzes (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[20 (Q) + 40 (T) + 40 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & each Quiz will be evaluated for 10 marks, and Final Quiz marks adding up to 20 marks. THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.	**20**
2.	**TESTS:** Students will be evaluated in a test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS**. **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: III						
Course Code	:	MCA263D4	**PRINCIPLES OF UI/UX DESIGN**	CIE Marks	:	100
Credits L-T-P	:	3-0-0	*(Theory)*	SEE Marks	:	100
Hours	:	45L	(Professional Elective Course) -D	SEE Duration	:	3 Hours

UNIT - I	**9 Hours**

Elements of UX Design-I- Introduction, from product design to user experience design, designing for experience, User experience and the web, Building from bottom to top.
Strategy Plane: Product Objectives, Business goals, Brand Identity, Success Metrics and User Needs, User Segmentation, Usability and User Research, Creating Personas
Scope Plane: Defining the Scope, Functional specifications, Content requirements.

UNIT - II	**9 Hours**

Elements of UX Design -II
Structure Plane: Interaction Design, Conceptual Models, Error Handling, Information Architecture
Skeleton Plane: Interface Design, Navigation Design, Information Design, Wireframes
Surface Plane: Sensory Design, Making Sense of the Senses, Contrast and Uniformity, Internal and External Consistency, Color Palettes and Typography, Design Comps and Style Guides.

UNIT - III	**9 Hours**

UI Design Process
Usability of Interactive Systems: Introduction, Usability Goals and Measures, Usability Motivation, Universal Usability, Guideline, principles, and theories
Managing Design Processes: Introduction, Organizational Design to support Usability, The Four Pillars of Design, Development methodologies, Ethnographic Observation, Participatory Design, Scenario Development

UNIT - IV	**9 Hours**

User Interface Evaluation and Interacting Styles
Evaluating Interface Design: Introduction, Expert Reviews, Usability Testing and Laboratories, Survey Instruments, Acceptance tests, Evaluation during Active Use, Controlled Psychologically Oriented Experiments.
Menu Selection, Form Filling and Dialog Boxes: Introduction, Task-Related Menu Organization, Single Menus, Combination of Multiple Menus, Content Organization, Fast Movement Through, Menus, Data Entry with Menus, Form Filling, Dialog Boxes and Alternatives, Audio Menus and Menus for Small Displays

UNIT - V	**9 Hours**

Patterns For Effective Interaction Design
Using social media: The Patterns-Editorial Mix, Personal Voices, Repost and Comment, Inverted Nano-pyramid, Sharing Widget, Content Leaderboard.
Going Mobile: Patterns-Vertical Stack, Touch Tools, Bottom Navigation, Thumbnail-and-Text List, Loading Indicators, Richly Connected Apps.
Visual Style and Aesthetics: Visual Design for Desktop applications, The Patterns Deep Background, Few Hues Many Values, Contrasting Font Weights Skins and Themes
Case Study: To explore the UI/UX using Digital tools, dark Patterns

Course Outcomes:
After going through this course the student will be able to:

CO1	:	Apply the theoretical foundations and awareness of User Interface and User Experience Design
CO2	:	Design based on the knowledge of features, approach, and patterns for designing UI and UX for cross platform applications
CO3	:	Identify and Apply various Design Skills in UI and UX for real world Applications
CO4	:	Evaluate UI/UX design Process/ artifacts for building products

Reference Books
1. Jesse James Garrett, "The Elements of User Experience: User-Centered Design for the Web and Beyond", New Riders Publishers ,2nd Edition, 2011, ISBN-13: 978-0321683687
2. Ben Shneiderman, Plaisant, Cohen, Jacobs, "Designing the User Interface", Pearson Education, 5th Edition, 2014, ISBN-10: 9332518734 ISBN-13: 978-9332518735
3. Bill Buxton, "Sketching User Experiences: Getting the Design Right and the Right Design", Morgan Kaufmann, 1st Edition,2007, ISBN-10: 0123740371 ISBN-13: 978-0123740373
4. Jenifer Tidwell, Charles Brewer, and Aynne Valencia, "Designing Interfaces- Patterns for Effective Interaction Design", O'Reilly®, 3rd Edition, ISBN 978-1492051916
5. Jeff Gothelf and Josh Seiden, "Lean UX: Applying Lean Principles to Improve User Experience", O'Reilly Media , 3rd Edition , ISBN 978-1492080359,2021
E-Resources https://maze.co/collections/ux-ui-design/tools/#list https://careerfoundry.com/en/blog/ux-design/dark-patterns-ux/

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of TWO Quizzes (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[20 (Q) + 40 (T) + 40 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZZES:** Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted & each Quiz will be evaluated for 10 marks, and Final Quiz marks adding up to 20 marks. THE SUM OF TWO QUIZZES WILL BE CONSIDERED AS FINAL QUIZ MARKS.	**20**
2.	**TESTS:** Students will be evaluated in a test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). TWO TESTS will be conducted. Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS**. **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (20) & Phase II (20) ADDING UPTO 40 MARKS.**	**40**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: III						
Course Code	:	MCA461P	**MINOR PROJECT**	CIE Marks	:	50
Credits: L-T-P	:	0-0-3	*(Practice)*	SEE Marks	:	50
Hours	:	06		SEE Duration	:	3 Hours

GUIDELINES

1. Student can form group of two to execute the Minor Project.
2. The student shall undertake a minor project aligned with their chosen electives, and it may be research-based or industry-oriented.
3. Allocation of the guides preferably in accordance with the expertise of the faculty.
4. Minor project topics could be aligned to any of the Centre of Excellence (CoE)/ Center of Competence (CoC) domain. The details of these could be obtained by visiting the website https://rvce.edu.in/rvce-center-excellence.
5. Minor project has to be implemented/executed in-house, using the resources available in the department/college/CoE/CoC.
6. Students have to note the periodic progress in the Minor Project Diary and report the work carried to their respective guides.
7. Students have to present the Minor project work to the departmental committee and only upon approval by the committee, the student can proceed to prepare and submit the hard copy of the final Minor project report.

The reports shall be printed on A4-size paper, with 1.5 line spacing, in Times New Roman font, size 12 and the report shall be softbound.

Course Outcomes:
After going through this course, the students will be able to

CO1	Analyze the research gaps, formulate the problem definition, conceptualize the objectives and design solution to cater to specific problems.
CO2	Apply higher order thinking skills and develop skill competencies specific to program specialization to implement real world problems with professional ethical standards.
CO3	Demonstrate the skill and knowledge by applying appropriate tools and techniques specific to their domain.
CO4	Communicate, work in teams and demonstrate the learning through oral presentations and report writing.

The evaluation criteria shall be as per the rubrics given below:

Phase	Activity	Weightage
I	Approval of the selected topic, formulation of Problem Statement and Objectives along with Synopsis submission	10%
II	Demonstrate the skill and knowledge by applying appropriate tools/techniques to design solution specific to the problem.	30%
III	Demonstrates the work carried out through experimental results, analysis and testing. Exhibits writing and communication skills through presentations, report writing and paper publication.	60%

Scheme for Semester End Evaluation (SEE):

The SEE examination shall be conducted by an external examiner (domain expert) and an internal examiner. Evaluation shall be done in batches, not exceeding 6 students per batch.

Sl. No	Contents	Marks
1	Write Up	20%
2	Demonstration of Project Work	60%
3	Viva Voce	20%

SEMESTER: III						
Course Code	:	MCA462N	**INTERNSHIP**	CIE Marks	:	100
Credits: L-T-P	:	0-0-6	*(Practice)*	SEE Marks	:	100
Hours	:	12		SEE Duration	:	3 Hours

GUIDELINES

1.Students can opt for undergoing internship at the industry or research organizations like BEL, DRDO, ISRO, NAL, etc.

2. Students must submit letter from the industry/research organizations, clearly specifying the candidate's name and the duration of the internship on the company letter head with authorized signature.

3. The duration of the internship shall be for a period of 6 weeks on full time basis after II semester final exams and before the commencement of III semester.

4. Students can approach the Centre of Excellence (CoE) in various domains and Center of Competence (CoC) hosted by RVCE for registering and working on relevant domain for training/internship. The details of these could be obtained by visiting the website https://rvce.edu.in/rvce-center-excellence.

5. Internship must be related to the field of specialization of the respective PG program in which the student has enrolled.

6. Students undergoing internship training are advised to report their progress and submit periodic progress reports/diary to their respective guides.

7. Students have to present the internship activities carried out to the departmental committee and only upon approval by the committee, the student can proceed to prepare and submit the hard copy of the final internship report.

The reports shall be printed on A4-size paper, with 1.5 line spacing, in Times New Roman font, size 12 and the report shall be softbound.

Course Outcomes: After going through this course, the student will be able to	
CO1	Explore the workplace, operating procedures of the department/company and its products, and other organizational concepts
CO2	Learn and improve writing and communication skills, research and technology, work in a team, and develop leadership skills
CO3	Apply higher order thinking skills - critical thinking, analysis, synthesis and evaluate complex problems to solve real world problems with professional ethical standards.
CO4	Develop and demonstrate skill competencies and knowledge specific to program specialization by applying appropriate tools and techniques.

Scheme of Continuous Internal Evaluation (CIE):
The evaluation committee shall consist of a Guide, Professor/Associate Professor and Assistant Professor. The committee shall assess the presentation and the progress reports in two reviews.

Reviews	Activity	Weightage
Review I	Ability to comprehend the functioning/operating procedures of the Organization/Departments. Application of Engineering knowledge, Critical thinking and analysis to solve problems.	40%
Review II	Demonstrates skill competencies, Resource Management and Sustainability. Exhibits writing and communication skills through presentations and report writing.	60%

Scheme for Semester End Evaluation (SEE): The SEE examination shall be conducted by an external examiner and an internal examiner. Evaluation shall be done in batches, not exceeding 6 students per batch.

Sl. No	Contents	Marks
1	Write Up	20%
2	Demonstration of Internship Work	60%
3	Viva Voce	20%